

## P2 Report

Reasoning:

The task requires choosing the most cost-efficient sorting algorithm based on different costs. Selection sort, Insertion sort and Bubble sort have different characteristics in terms of comparisons and swaps, so we need to estimate the cost of them before actual sorting. Accord to the material, we need only focus on the reads/writes performed on the input array.

The estimation uses a simplified cost model. It assumes worst-case behaviour.

For Selection sort,

We need to compare  $[(n-1) + (n-2) + (n-3) + \dots + 1] = (n-1) * n/2$  times and we need to swap at most  $(n-1)$  times.

Therefore,

```
int read_count = 2*(n-1) + n * (n - 1);  
int write_count = 2*(n-1);
```

For Insertion sort,

We need to compare  $[(n-1) + (n-2) + \dots + 1] = (n-1)*n/2$  times and we need to swap the same times as that of comparison.

Therefore,

```
int read_count = 2* n * (n-1);  
int write_count = n * (n-1);
```

For Bubble sort,

We need to compare  $(n-1)*n$  times and We need to swap  $[(n-1) + (n-2) + \dots + 1] = (n-1)*n/2$  times.

Therefore,

```
int read_count = 2*n*(n-1) + n * (n - 1);  
int write_count = n*(n-1);
```

After estimation of each sorting method, we could know which one is the most efficient way. And then choose the most efficient way to deal with this task.

Evidence of Testing:

```
The cost for Selection sort is: 126  
The cost for Insertion sort is: 270  
The cost for Bubble sort is: 360  
So I would choose: Selection sort  
Cost of sorting was 126  
# likes are:  
9 8 7 6 5 4 3 2 1 0
```

```
/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/P2/P2task  
The cost for Selection sort is: 126  
The cost for Insertion sort is: 270  
The cost for Bubble sort is: 360  
So I would choose: Selection sort  
Cost of sorting was 126  
# likes are:  
100 96 92 88 84 80 76 72 68 64
```

Word Count: 185.